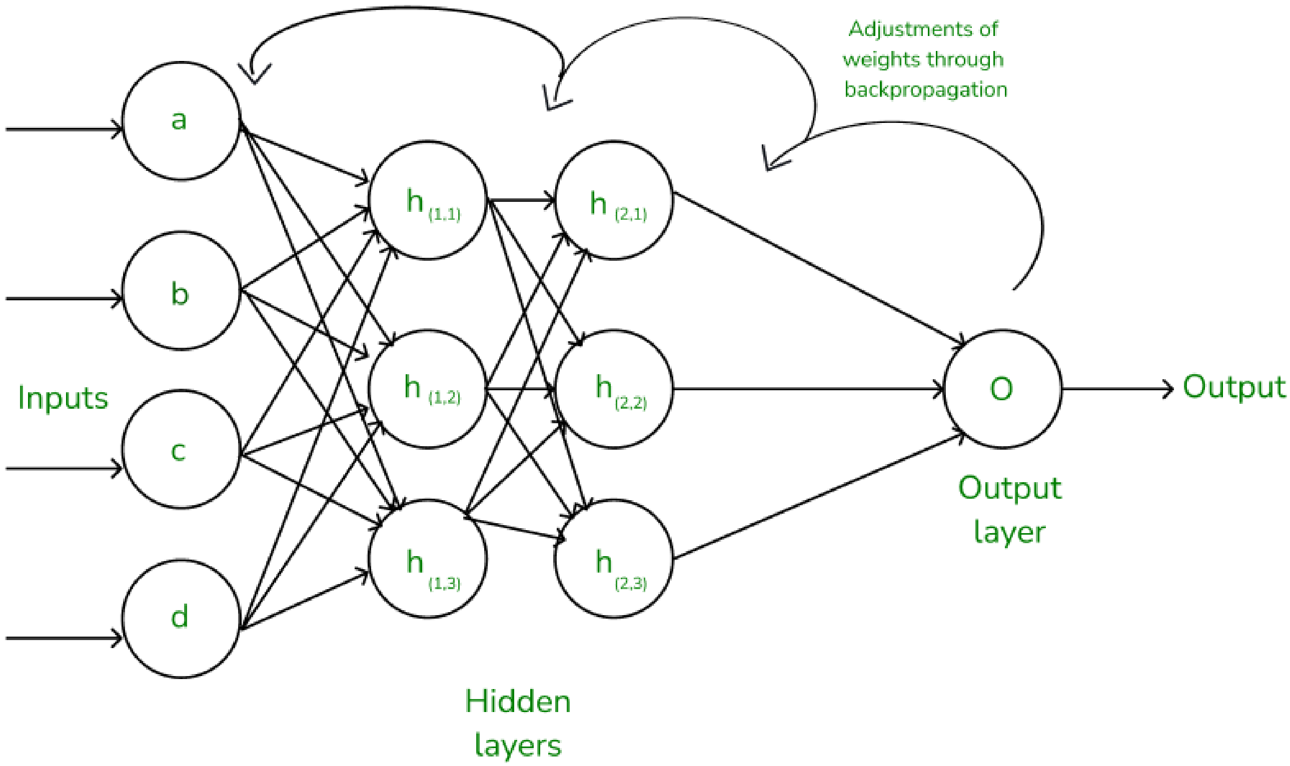
Backpropagation

[AIDL]

Backpropagation is a method that tells a neural network **how wrong it is** and adjusts the weights so the error becomes smaller next time.

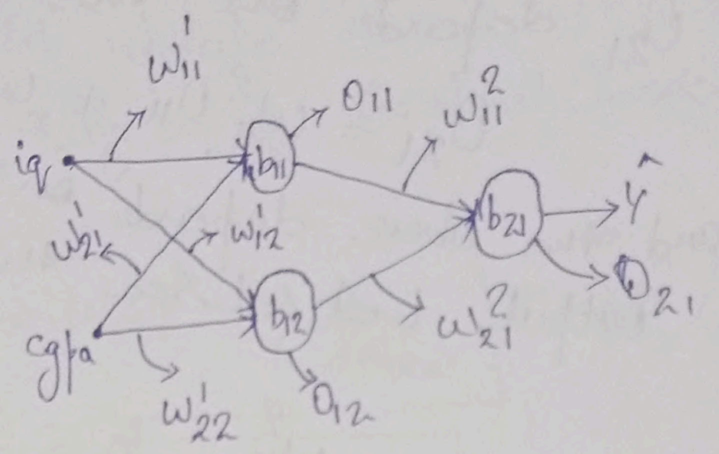
Why is it called “back-propagation”?

Because the **error is propagated backward** from the output layer → hidden layers → input layer to compute how each weight contributed to the error.



Backpropagation

iq	cgpa	lpa
80	8	3
60	9	5
70	5	8
120	7	11



We have taken activation → Linear

Step: 0) init^n w, b (random or w → 1, b → 0)

Step 1) You select a row (point) → sing student detail.

Step 2) ~~Pre~~ Forward Pass

* Input data flows through the network (input → hidden → output)

* Prediction ($\hat{y}$) is generated

Step 3) choose a loss fuction | Loss Calculation.

* compare $\hat{y}$ and y

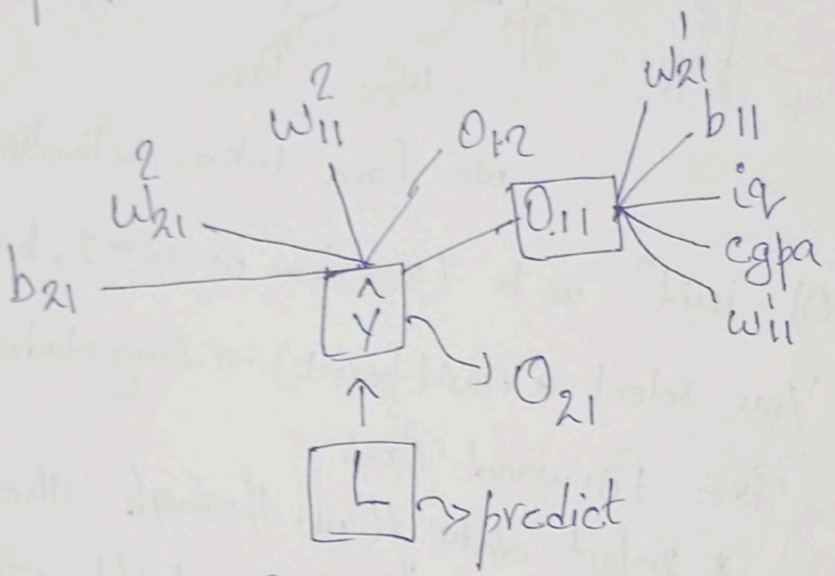
* loss function (MSE, ~~...~~ MAE)

Note:.
Let we get some loss, that loss depend ~~on the~~ output O_{21} and that

O_{21} depends on the

$$O_{21} = w_{11}^2 O_{11} + w_{21}^2 O_{12} + b_{21}$$

and the above depends on the ~~ant~~ other output, let ~~set~~ see the flow.



At above we had seen that the final predict is depends on the output of the various nodes or weigh & bias so we have to update in backpropagation.

<u>Step 4)</u> Backward pass (Backpropagation)

* Compute the gradients
* propagate error from output layer back to input layer.

Step 5 weight update

* By using Gradiant Descent

$$\boxed{W_{new} = W_{old} - \eta \frac{\partial L}{\partial W_{old}}} \rightarrow eq\ X$$

$$\boxed{b_{new} = b_{old} - \eta \frac{\partial L}{\partial b_{old}}} \rightarrow eq\ Y$$

So, Similarly

slope / loss derivative / partial

$$w^{2}_{11\,new} = w^{2}_{11\,old} - \eta \frac{\partial L}{\partial w^{2}_{21\,old}}$$

$$w^{2}_{21\,new} = w^{1}_{21\,old} - \eta \frac{\partial L}{\partial w^{2}_{21\,old}}$$

$$b_{21\,new} = b_{21\,old} - \eta \frac{\partial L}{\partial b_{21\,old}}$$

So, by above network / model we have to calculate 9 parameters.

$$\frac{\partial L}{\partial w^{2}_{11}}, \frac{\partial L}{\partial w^{2}_{21}}, \frac{\partial L}{\partial b_{21}} \Bigg| \frac{\partial L}{\partial w^{1}_{11}}, \frac{\partial L}{\partial w^{1}_{21}}, \frac{\partial L}{\partial b_{11}} \Bigg| \frac{\partial L}{\partial w^{1}_{12}}, \frac{\partial L}{\partial w^{1}_{22}}$$

$\frac{\partial L}{\partial b_{12}}$

$\frac{\partial L}{\partial b}$

$$\frac{\partial L}{\partial w_{11}^2} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial w_{11}^2}$$

→ Chain rule of diff^n

now,

$$\frac{\partial L}{\partial \hat{y}} = \frac{\partial}{\partial \hat{y}} (y-\hat{y})^2 \Rightarrow -2(y-\hat{y})$$

$$\frac{\partial \hat{y}}{\partial w_{11}^2} = \frac{\partial}{\partial w_{11}^2} \left[O_{11} w_{11}^2 + O_{12} w_{21}^2 + b_{21} \right]$$

$$= O_{11}$$

now,

$$\boxed{\frac{\partial L}{\partial w_{11}^2} = -2(y-\hat{y}) O_{11}}$$

Similarly we have to compute for another 8 eq.

or we

$$\frac{\partial L}{\partial w_{21}^2} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial w_{21}^2}$$

$\longrightarrow$ again apply chain rule.

$\frac{\partial L}{\partial \hat{y}} \Rightarrow$ already compute $\Rightarrow -2(y-\hat{y})$

$$\frac{\partial \hat{y}}{\partial w_{21}^2} = \frac{\partial}{\partial w_{21}^2}\left[O_{11}w_{11}^2 + w_{21}^2 O_{21} + b_{21}\right]$$

$$= O_{12}$$

$$\boxed{\frac{\partial L}{\partial w_{21}^2} \Rightarrow -2(y-\hat{y})\, O_{12}}$$

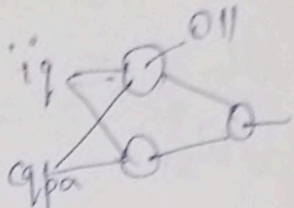
$$\frac{\partial L}{\partial b_{21}} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial b_{21}}$$

$$\frac{\partial L}{\partial \hat{y}} = -2[y-\hat{y}]$$

$$\frac{\partial \hat{y}}{\partial b_{21}} = 1$$

$$\frac{\partial L}{\partial b_{21}} = -2(y-\hat{y})$$

Now, we go for $\underline{\text{output } O_{11}}$



for, $$\frac{\partial L}{\partial w_{11}^1} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial O_{11}} \frac{\partial O_{11}}{\partial w_{11}^1}$$

$= \frac{\partial L}{\partial \hat{y}}$ is already define &

$$\frac{\partial \hat{y}}{\partial O_{11}} = \frac{\partial}{\partial O_{11}}[w_{11}^2 O_{11} + w_{21}^2 O_{21} + b_{21}]$$

$$= w_{11}^2$$

$$\frac{\partial O_{11}}{\partial w_{11}^1} = \frac{\partial}{\partial w_{11}^1}(iq\, w_{11}^1 + cgpa\, w_{21}^1 + b_{11})$$

$$= \boxed{iq \;\&\; x_{i1}}$$

So,

$$\boxed{\frac{\partial L}{\partial w_{11}^1} = -2(y-\hat{y})\, w_{11}^2\, x_{i1}}$$

now,

$$\frac{\partial L}{\partial w_{21}^1} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial O_{11}} \frac{\partial O_{11}}{\partial w_{21}^1}$$

$\frac{\partial L}{\partial \hat{y}}$ already computed

$$\frac{\partial \hat{y}}{\partial O_{11}} = \frac{\partial}{\partial O_{11}}[w_{11}^2 O_{11} + w_{21}^2 O_{21} + b_{21}] \Rightarrow w_{11}^2$$

$$\frac{\partial O_{11}}{\partial w^1_{21}} = \frac{\partial}{\partial w^1_{21}} \left(iq\, w^1_{11} + cgpa\, w^1_{21} + b_{11} \right)$$

$$= cgpa \rightarrow x_{i2}$$

So, →

$$\boxed{\frac{\partial L}{\partial w^1_{21}} = -2(y-\hat{y})\, w^2_{11}\, x_{i2}}$$

Now,

$$\frac{\partial L}{\partial b_{11}} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial O_{11}} \frac{\partial O_{11}}{\partial b_{11}}$$

$$= -2(y-\hat{y})\, w^2_{11} \quad 1$$

$$\boxed{\frac{\partial L}{\partial b_{11}} = -2(y-\hat{y})\, w^2_{11}}$$

Now we go $\boxed{\text{for } O_{12}}$ → node ②

$$\frac{\partial L}{\partial w^1_{12}} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial O_{12}} \frac{\partial O_{12}}{\partial w^1_{12}}$$

$$= -2(y-\hat{y}) \cdot$$

$$\frac{\partial \hat{y}}{\partial O_{12}} = \frac{\partial}{\partial O_{12}} \left(O_{11} w^2_{11} + O_{12} w^2_{21} + b_{21} \right) \Rightarrow w^2_{21}$$

$$\frac{\partial O_{12}}{\partial w_{12}} = \frac{\partial [w_{12}\, iq + cgpa\, w'_{22} + b_{21}]}{\partial w_{12}}$$

$$= cgp \cdot iq \rightarrow x_{i1}$$

So,

$$\boxed{\frac{\partial L}{\partial w'_{12}} = -2 \cdot (y - \hat{y})\, w^2_{21}\, x_{i1}}$$

Similarly,

$$\frac{\partial L}{\partial w'_{22}} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial O_{12}} \frac{\partial O_{12}}{\partial w'_{22}}$$

$$\boxed{= -2(y - \hat{y})\, w^2_{21}\, x_{i2}}$$

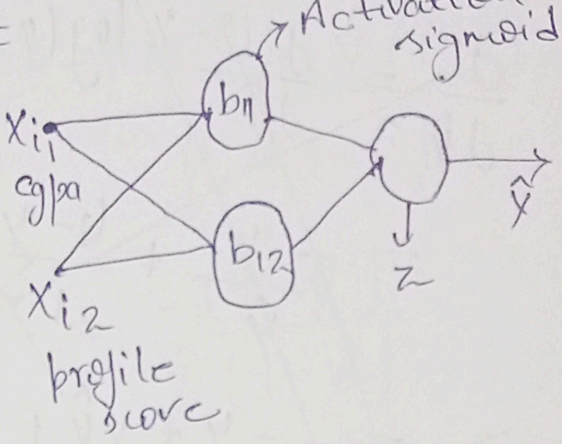
$$\frac{\partial L}{\partial b_{12}} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial O_{12}} \frac{\partial O_{12}}{\partial b_{12}}$$

$$= -2(y - \hat{y})\, w'_{21} \cdot 1$$

$$\boxed{\frac{\partial L}{\partial b_{12}} = -2(y - \hat{y})\, w'_{21}}$$

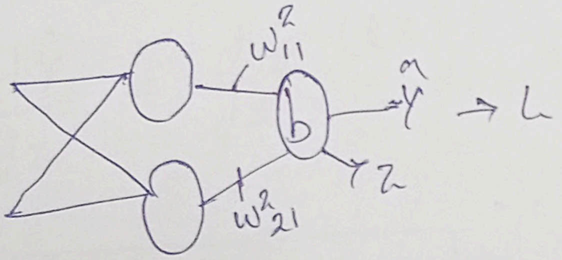
Now, we go for classification Example

cgpa	profile score	placement
8	8	1
7	9	1
6	10	0
5	5	0



We have total 9 trainable parameter 6w 3b

$$\text{Loss} = -y\log(\hat{y}) - (1-y)\log(1-\hat{y})$$



Note:

$\boxed{\hat{y} = \sigma(z) \text{ / or } \sigma(O_{21})}$

$$\boxed{\frac{\partial L}{\partial w_{11}^2} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z} \frac{\partial z}{\partial w_{11}^2} \Rightarrow -(y-\hat{y})O_{11}}$$

$$\boxed{\frac{\partial L}{\partial w_{21}^2} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z} \frac{\partial z}{\partial w_{21}^2} \Rightarrow -(y-\hat{y})O_{12}}$$

$$\boxed{\frac{\partial L}{\partial b_{21}} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z} \frac{\partial z}{\partial b_{21}} \Rightarrow -(y-\hat{y})}$$

These equation are solved below.

$$\frac{dL}{d\hat{y}} = \frac{d}{d\hat{y}}[-y\log(\hat{y}) - (1-y)\log(1-\hat{y})]$$

$$= \frac{-y}{\hat{y}} + \frac{(1-y)}{(1-\hat{y})} = \frac{-y(1-\hat{y}) + \hat{y}(1-y)}{\hat{y}(1-\hat{y})}$$

$$= \frac{-y + \cancel{y\hat{y}} + \hat{y} - \cancel{y\hat{y}}}{\hat{y}(1-\hat{y})} = -\frac{(y-\hat{y})}{\hat{y}(1-\hat{y})}$$

$$\boxed{\frac{dL}{d\hat{y}} = \frac{-(y-\hat{y})}{\hat{y}(1-\hat{y})}}$$

$$\frac{d\hat{y}}{dz} = \frac{d(\sigma(z))}{dz} \Rightarrow \sigma(z)[1-\sigma(z)]$$

$$= \hat{y}(1-\hat{y})$$

$$\boxed{\frac{d\hat{y}}{dz} = \hat{y}(1-\hat{y})}$$

<u>LOOK</u> we have to compute/Mult" $\frac{d\hat{y}}{dz} \frac{dL}{d\hat{y}}$

$\hat{y}\cancel{(1-\hat{y})}$

So, $\frac{dL}{d\hat{y}}\frac{d\hat{y}}{dz} = \frac{-(y-\hat{y})}{\hat{y}\cancel{(1-\hat{y})}}$

So, $\boxed{\frac{dL}{d\hat{y}}\frac{d\hat{y}}{dz} = -(y-\hat{y})}$

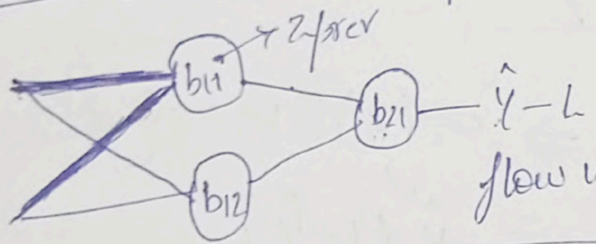
$$\frac{\partial z}{\partial w_{11}^2} = \frac{\partial (w_{11}^2 O_{11} + w_{21}^2 O_{12} + b_{21})}{\partial w_{11}^2} = O_{11}$$

$$\frac{\partial z}{\partial w_{21}^2} = \frac{\partial (w_{11}^2 O_{11} + w_{21}^2 O_{12} + b_{21})}{\partial w_{21}^2} = O_{12}$$

Similar $\frac{\partial z}{\partial b_{21}} = 1$

Now we will compute:-

$\frac{\partial L}{\partial w_{11}^1}$, $\frac{\partial L}{\partial w_{21}^1}$, $\frac{\partial L}{\partial b_{11}}$



flow will be $L - \hat{y} - z_f \rightarrow O_{11}$

$w_{11}^1 \leftarrow z_{prev} \leftarrow$

$$\frac{\partial L}{\partial w_{11}^1} = \boxed{\frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z}} \frac{\partial z}{\partial O_{11}} \frac{\partial O_{11}}{\partial z_{prev}} \frac{\partial z_{prev}}{\partial w_{11}^1}$$

$$= -(y - \hat{y}) \quad w_{11}^2 \quad O_{11}(1 - O_{11}) \quad X_{i1}$$

$$\frac{\partial z}{\partial o_{11}} = \frac{\partial (w_{11}^2 O_{11} + w_{21}^2 O_{12} + b_{21})}{\partial o_{11}} = \boxed{w_{11}^2}$$

$$\frac{\partial O_{11}}{\partial z_{prev}} = \sigma(z_{prev}) \Rightarrow \sigma(z_{prev})[1 - \sigma(z_{prev})]$$

$$\boxed{\Rightarrow O_{11}(1 - O_{11})}$$

$$\frac{dz_{prev}}{dw_{11}^1} = \frac{d}{dw_{11}^1}\left(w_{11}^1 x_{i1} + w_{21}^1 x_{i2} + b_{11}\right)$$

$$= x_{i1}$$

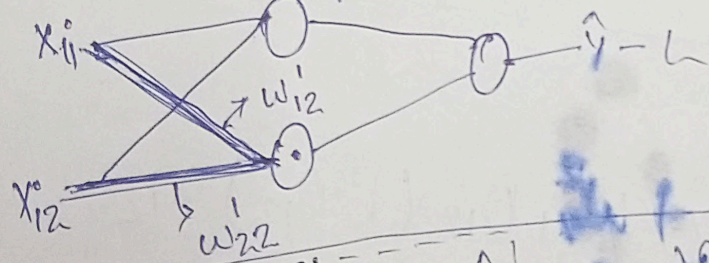
$$\boxed{\frac{dL}{dw_{11}^1} = -(y-\hat{y})\, w_{11}^2\, O_{11}\,(1-O_{11})\, x_{i1}}$$

Similar

$$\boxed{\frac{dL}{dw_{21}^1} = -(y-\hat{y})\, w_{11}^2\, O_{11}\,(1-O_{11})\, x_{i2}}$$

$$\boxed{\frac{dL}{db_{11}} = -(y-\hat{y})\, w_{11}^2\, O_{11}\,(1-O_{11})\cdot 1}$$

Now we compute $\frac{dL}{dw_{12}}, \frac{dL}{dw_{22}^1}, \frac{dL}{db_{12}}$



$$\boxed{\begin{aligned}\frac{dL}{dw_{12}^1} &= \frac{dL}{d\hat{y}}\,\frac{d\hat{y}}{dz}\,\frac{dz}{dO_{12}}\,\frac{dO_{12}}{dz_{prev}}\,\frac{dz_{prev}}{dw_{12}} \\ &= -(y-\hat{y})\quad w_{21}^2\quad O_{12}(1-O_{12})\quad x_{i1}\end{aligned}}$$

These equation are computed below.

$$\frac{\partial z}{\partial O_{12}} = \frac{\partial}{\partial O_{12}} (w^2_{11} O_{11} + w^2_{21} O_{12} + b_{21})$$

$$\boxed{= w^2_{21}}$$

$$\frac{\partial O_{12}}{\partial z_{prev}} = \sigma(z_{prev}) \Rightarrow \sigma(z_{prev})[1 - \sigma(z_{prev})]$$

$$\boxed{= O_{12}[1 - O_{12}]}$$

$$\cancel{\frac{\partial z_p}{\partial w^1_{12}} = \frac{\partial}{\partial w^1_{12}} (w^1_{12} \cdot x}$$

$$\frac{\partial z_{prev}}{\partial w^1_{12}} = \frac{\partial}{\partial w^1_{12}} (w^1_{12} x_{i1} + w^1_{22} x_{i2} + b_{12}) = x_{i1}$$

$$\boxed{\frac{\partial L}{\partial w^1_{12}} = -(y - \hat{y})\, w^2_{21}\, O_{12}\, (1 - O_{12})\, x_{i1}}$$

Similarly.

$$\boxed{\frac{\partial L}{\partial w^2_{22}} = -(y - \hat{y})\, w^2_{21}\, O_{12}\, (1 - O_{12})\, x_{i2}}$$

$$\boxed{\frac{\partial L}{\partial b_{12}} = -(y - \hat{y})\, w^2_{21}\, O_{12}\, (1 - O_{12}) \cdot 1}$$